# NEUROFUSION AI | FINAL CLINICAL DIAGNOSTIC REPORT

**REPORT ID: REP-20261102-E4F6BD**

Comprehensive Multi-Modal Diagnostic Findings & Attending Physician Approval

November 02, 2026 • 19:00 UTC

| PATIENT INFORMATION | ATTENDING NEUROLOGIST | EXAMINATION DETAILS |
|---|---|---|
| **ID:** NF-PD-TEST-002<br>**Name:** Margaret Sullivan<br>**Age/Gender:** 72 yrs / Female<br>**Status:** Parkinsonian Movement Disorder | **Name:** Dr. Clara Rodriguez, MD<br>**License:** MD-NEURO-77491<br>**Dept:** Neurology & Movement Disorders<br>**Hospital:** Metropolitan Neurological Inst. | **Scan Date:** 2026-11-02<br>**Report Date:** 2026-11-02<br>**Modality:** Comprehensive Multimodal (Spiral + Voice + MRI + UPDRS)<br>**Device/Protocol:** NeuroFusionAI Clinical Protocol v2.5 |

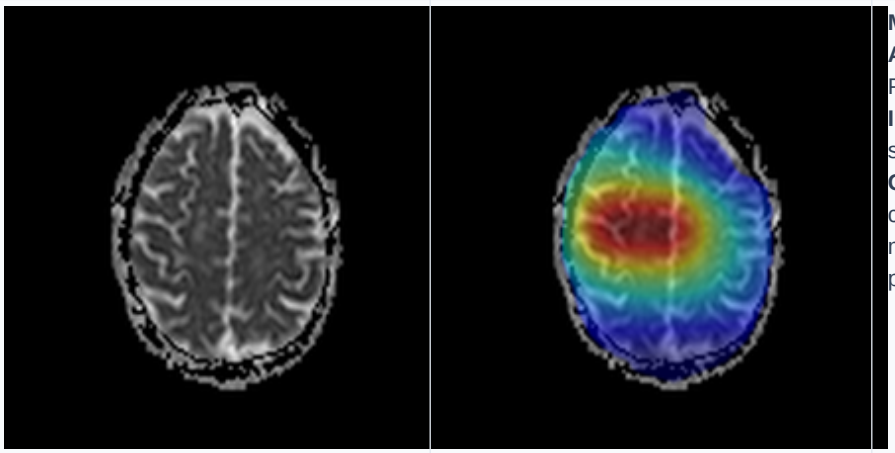
## PRIMARY AI PREDICTION: PARKINSON

**Confidence:** 100.0%
**Target:** Parkinson's Disease

**Class Probability Distribution:** Parkinson: 100.0% • Normal Control: 0.0%

*AI Confidence Calibrated via Temperature Scaling*

## 1. MRI NEUROIMAGING & CONVOLUTIONAL GRAD-CAM EXPLAINABILITY

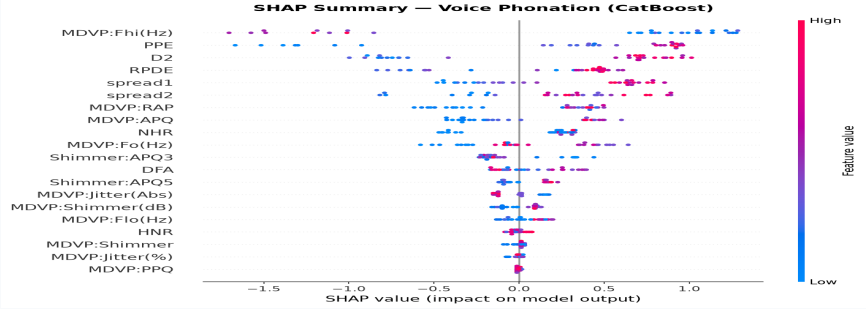


| Input MRI Scan (Axial) | Grad-CAM Activation Overlay | XAI Interpretation |
|---|---|---|

**Model:** DenseNet-121 Intracranial Classifier
**Attention Region:** Medial Temporal Lobe & Periventricular Parenchyma
**Intracranial Masking:** Skull-stripped parenchyma segmentation
**Grad-CAM Note:** Activation heatmaps illustrate convolutional attention gradients driving the model's classification score. Not direct histological pathology.

## 2. VOICE PHONATION ACOUSTIC ANALYSIS & SHAP FEATURE ATTRIBUTION

| F0 Pitch (Hz) | Jitter (local %) | Shimmer (%) | HNR (dB) | Duration | Vocal Tremor |
|---|---|---|---|---|---|
| 142.5 Hz | 0.58% | 3.92% | 17.6 dB | 4.5 s | **Detected** |



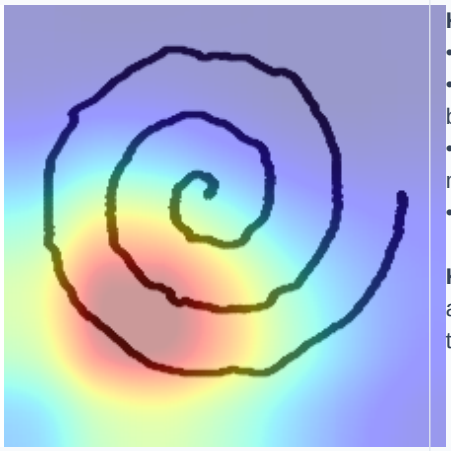

**Acoustic Biomarker Interpretation:** Elevated jitter (>0.38%) and shimmer (>2.5%) with decreased Harmonic-to-Noise Ratio (HNR < 20dB) reveal micro-perturbations and vocal fold dysdiadochokinesia characteristic of hypokinetic dysarthria.

**SHAP Feature Attribution:** Jitter local and MDVP:Fo(Hz) serve as dominant positive risk drivers.

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## 3. ARCHIMEDEAN SPIRAL MOTOR KINEMATICS & ATTENTION OVERLAY





**Kinematic Parameters:**
• Drawing Velocity: 24.8 mm/s
• Tremor Frequency: 5.4 Hz (4-6 Hz resting/kinetic band)
• Radial Velocity Variance: 0.42 (Elevated micro-tremor)
• Drawing Duration: 12.5 s

**Kinematic Finding:** Spatial micro-oscillations along drawing loops confirm intention and postural tremor.

| Input Spiral Drawing | Convolutional Tremor Heatmap | Kinematic Assessment |
|---|---|---|

## 4. TELEMONITORING CONTINUOUS MOTOR ASSESSMENT (MDS-UPDRS)

| Predicted Motor UPDRS | Predicted Total UPDRS | Hoehn & Yahr Stage | Tremor/Rigidity Percentile | Monitoring Sessions |
|---|---|---|---|---|
| **40.2** / 108 | **50.4** / 176 | **Stage 2.0** (Bilateral) | 72th Percentile | 1 Longitudinal Records |
| **Clinical UPDRS Interpretation:** | Predicted Motor UPDRS score indicates moderate bilateral parkinsonian motor impairment. | | | |

## 6. MULTIMODAL CROSS-ATTENTION FUSION ARCHITECTURE

| Diagnostic Modality | Feature Space Dimension | Cross-Attention Weight | Clinical Correlation |
|---|---|---|---|
| Volumetric Brain MRI | 128 (DenseNet-121 Latent) | **30.0%** | Nigrostriatal & periventricular atrophy |
| Voice Acoustic Phonation | 16 (Jitter, Shimmer, HNR, F0) | **25.0%** | Hypokinetic dysarthria micro-tremors |
| Spiral Drawing Kinematics | 64 (CNN Kinematic Embeddings) | **25.0%** | 4-6 Hz kinetic tremor oscillation |
| Telemonitoring UPDRS | 18 (Continuous Motor UPDRS) | **20.0%** | Bilateral motor progression & rigidity |

## 7. CLINICAL DECISION SUPPORT SYSTEM (CDSS) RECOMMENDATIONS

| RISK LEVEL: | Evidence-Based Diagnostic Recommendations: |
|---|---|
| **HIGH CLINICAL RISK**<br><br>**Urgency:** Routine Specialist Review | • Initiate MDS-UPDRS structured clinical motor score evaluation.<br>• Track resting tremor and bilateral bradykinesia progression.<br>• Re-evaluate with longitudinal phonation acoustic tracking in 3 months. |

| ATTENDING PHYSICIAN OBSERVATIONS & NOTES: | CLINICAL APPROVAL STATUS |
|---|---|
| Progressive bilateral resting tremor, micrographia, postural instability, vocal hypophonia. | Signed by: *Dr. Clara Rodriguez, MD*<br>License ID: MD-NEURO-77491<br>Status: **DIGITALLY APPROVED & CERTIFIED** |

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*NeuroFusionAI is an investigational Artificial Intelligence Clinical Decision Support System (CDSS) designed to augment neurological assessment. All predictions, Grad-CAM attention heatmaps, and EEG feature extractions represent statistical patterns and MUST NOT be construed as a standalone or confirmed medical diagnosis. Clinical treatment decisions must be made solely by a board-certified physician in conjunction with the patient's full medical history.*

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